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15 MARCH 1982
SUPERSEDING
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15 DECEMBER 1976

MILITARY SPECIFICATION

AMPLIFIER, DOPPLER: ASSEMBLY 11740261

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, quality assurance provisions, and packaging for a doppler amplifier assembly, used in Radar Receiver R-1475/VPS-2, part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926 -	Fire Control Material: General Specification Governing the Manufacture and Inspection of
MIL-P-14232 -	Parts, Equipment and Tools for Army Material; Packaging and Packing of
MIL-I-45607 -	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-46357 -	Computer, Gun Direction; M18 Manufacturing Processes and Inspection

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

D11740261 Doppler Amplifier

PACKAGING DATA SHEET

D11740261 Packaging of Dopple: Amplifier

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Fabrication. The doppler amplifier shall be manufactured in accordance with Drawing 11740261 and drawings pertaining thereto.

3.2 General Specifications. Where applicable, the doppler amplifier shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings

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d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

3.2.2 MIL-C-46357.

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration. The doppler amplifier shall be mounted on the vibration table in normal service position. The doppler amplifier shall meet the requirements of 3.5 after being subject to continuous vibration for 2 hours along each of three mutually perpendicular axes within the frequency range of 10 to 55 hertz (Hz) and a half amplitude excursion of 1/64 inch. The rate of change of frequency at each frequency range shall not exceed 1 Hz per second. Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock. The doppler amplifier shall not suffer damage and shall meet the requirements of 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G) and a pulse duration of 11 $\pm$1 milliseconds (ms). The G value of each impulse shall be within $\pm$10 percent when measured with 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5.5 ms.

3.3.3 Altitude.

3.3.3.1 Operating. The doppler amplifier shall meet the requirements of 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Nonoperating. The doppler amplifier shall meet the requirements of 3.5 after exposure for not less than 1 hour to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The doppler amplifier shall meet the requirements of 3.5 when stabilized thermally at any temperature between -40°F and +155°F.

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3.3.4.2 Nonoperating (storage). The doppler amplifier shall not suffer permanent damage when subjected to not less than 4 hours at each temperature extreme of -65°F and +160°F, and shall meet the requirements of 3.5 when returned to room ambient.

3.3.5 Humidity (nonoperating). The doppler amplifier shall meet the requirements of 3.5 after exposure to an atmosphere with a relative humidity of 95 ±5 percent for a period of not less than 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the doppler amplifier.

3.5 Performance.

3.5.1 Input Voltage. The input voltage to the doppler amplifier shall be 25.0 ±0.1 volts direct current (Vdc) at 50 milliamperes (ma) applied to A1J1-1 (+) and A1J1-2 (ret) also A2J1-1 (+) and A2J1-2 (ret).

3.5.2 Output Noise. With the input terminals E1 and E2 of both A1 and A2 open circuited the output noise measured at A1J1-A1 and A2J1-A1 shall be 50 millivolts (mV) peak-to-peak (P-P) maximum, excluding any 60 Hz noise.

3.5.3 Gain. The gain of each amplifier A1 and A2 shall be 79.0 ±1.5 dB when a 1 KHz signal is applied to terminals E1 and E2 (ret) at a level sufficient to produce 300.0 ±0.1 mV at the output J1-A1.

3.5.4 Frequency Response. The frequency response of each amplifier A1 and A2 shall be as follows:

a. With the input level held constant at that level required to meet the condition of 3.5.3, and the input frequency varied between 900 Hz and 10 KHz, the output level measured at J1-A1 shall not vary more than a total of 1.0 dB.

b. When the input frequency is increased to 20 KHz the output level at J1-A1 shall be at least 2 dB below the highest output level observed between 900 Hz and 10 KHz.

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3.5.5 Limiting Action. The limiting action of each amplifier A1 and A2 shall be as follows:

a. When the input signal, at 1 KHz applied to terminals E1 and E2 (ret), has its level adjusted to just below output limiting (limiting threshold) the output voltage at J1-A1 shall be 1.5 ±0.5 volts P-P.

b. When the input level is increased by 20 dB and 40 dB above limiting threshold, the output level at J1-A1 shall not be greater than 2.0 volts P-P.

3.6 First Article Inspection. This specification makes provisions for First Article Inspection. Requirements for the submission of First Article Samples by the contractor shall be as specified in the contract.

3.7 Workmanship. All parts and assemblies shall be free from burrs, chips, sharp edges, cracks, unblended radii, surface blisters, excess flash, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Surface coating shall be continuous except for a few slight scratches not exposing the base material. The assembly operations shall be performed in a thorough, workmanlike manner consistent with the best practices of the industry. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

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4.2 First article (initial production) submission. The requirement for first article approval and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three doppler amplifiers manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witnes.. the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in the applicable Supplementary Quality Assurance Provision (SQAP). In the absence of SQAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in 4.3.2.3 and 4.4 herein. Examination and test for

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packaging and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in 4.3.2.3 and 4.4 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 Functional tests. The requirements and tests in Table I shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature (60°F to 90°F).

TABLE I. CLASSIFICATION OF DEFECTS

(100 percent inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR:		
101. Electrical	3.4	4.6.2
102. Performance	3.5	4.6.3
MINOR: NONE		

4.3.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling. One doppler amplifier shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and test in Table II. Three doppler amplifiers shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table III.

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4.6.1.4 Temperature.

4.6.1.4.1 High (Nonoperating and Operating). This test shall be performed as specified in Procedure I, Method 501 of MIL-STD-810. The temperature of Step 2 shall be +160°F. The temperature of Step 3 shall be maintaind for not less than 4 hours. The temperature of Step 4 shall be +155°F. In Step 5, the temperature shall be maintained for 4 hours and the doppler amplifier shall be tested in accordance with 4.6.3 during this time.

4.6.1.4.2 Low (Nonoperating and Operating). This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of Step 2 shall be -65°F maintained for not less than 4 hours. The temperature of Step 4 shall be -40°F. In Steps 5 and 7, the doppler amplifier shall be tested in accordance with 4.6.3.

4.6.1.5 Humidity (Nonoperating). This test shall be performed in accordance with Procedure II, Method 507 of MIL-STD-810. The doppler amplifier shall be tested in accordance with 4.6.3.

4.6.2 Electrical. Continuity shall be inspected by Standard Measuring Equipment (SME), and visual examination for compliance with 3.4.

4.6.3 Performance. The doppler amplifier shall be tested as follows: Set up the equipment as shown in Figure 1. Do not apply power to the doppler amplifier under test. All test equipment is to be connected to its 110 V, 60 Hz source through three-prong grounding plugs. All test equipment, except the oscilloscope, shall be placed on the brass test plate, Item 7. The oscilloscope must have a ground strap connected to the brass test plate.

4.6.3.1 Input Voltage. Apply the input voltage as specified in 3.5.1.

4.6.3.2 Output Noise. With no input signal applied to terminals E1 and E2 of either A1 or A2, use the oscilloscope, Item 5, to measure the output noise level as specified in 3.5.2.

4.6.3.3 Amplifier A1.

4.6.3.3.1 Gain. Adjust the attenuator, Item 3, for 80 decibels (dB) and connect the high side output to E1 and the low side to E2 of A1. Set the audio oscillator, Item 1, to 1 KHz and set its output level for a reading of 300.0 ±0.1 mV rms measured at output A1J1-A1,

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using the digital ac voltmeter, Item 6. Using the same ac voltmeter measure the signal level at the output of the audio oscillator, Item 1. It shall not be less than 300.0 mV rms and not more than 356.5 mV rms. Note this voltage.

TABLE IV. TEST EQUIPMENT LIST

ITEM NUMBER	DESCRIPTION	
1. Audio Oscillator	Frequency:	600 Hz to 20 KHz
	Amplitude:	10 to 400 mV variable
	Output Impedance:	50 ohms
	Max. Hum & Noise:	60 dB below rated output
2. Analog ac Voltmeter	Voltage Range:	to 400 mV
	Frequency Range:	600 Hz to 20 KHz min.
	Accuracy:	±1%
	Input Impedance:	10 megohms
3. Variable Attenuator	Attenuation:	10 dB to 80 dB in 5 dB steps
	Accuracy:	±0.5 dB of setting
	Impedance:	50 ohms
4. Power Supply	Output Voltage:	+25.0 ±0.1 Vdc
	Load Regulation:	0.01% max.
	Line Regulation:	0.01% max.
	Ripple & Noise:	200 uV max.
	Output Current:	50 mA min.
5. Oscilloscope	Bandwidth:	1 MHz min.
	Deflection Factor:	ac 10 mV/div to 1 V/div.
	Input Impedance:	1 megohm shunted by approximately 40 pF.
6. Digital ac Voltmeter	Frequency Range:	100 Hz to 20 KHz min.
	Voltage Range:	to 1 volt
	Accuracy:	±0.35% total
	Resolution:	10 uV
	Input Impedance:	1 megohm
7. Brass Test Plate	2 feet by 3 feet, approximately	
8. Termination	150 ohm, 1/2 W	
9. Termination	50 ohm, 1/2 W	
10. Connector (2)	DEM-5W1S, Cinch Mfg. Co., Chicago, Illinois or equivalent.	

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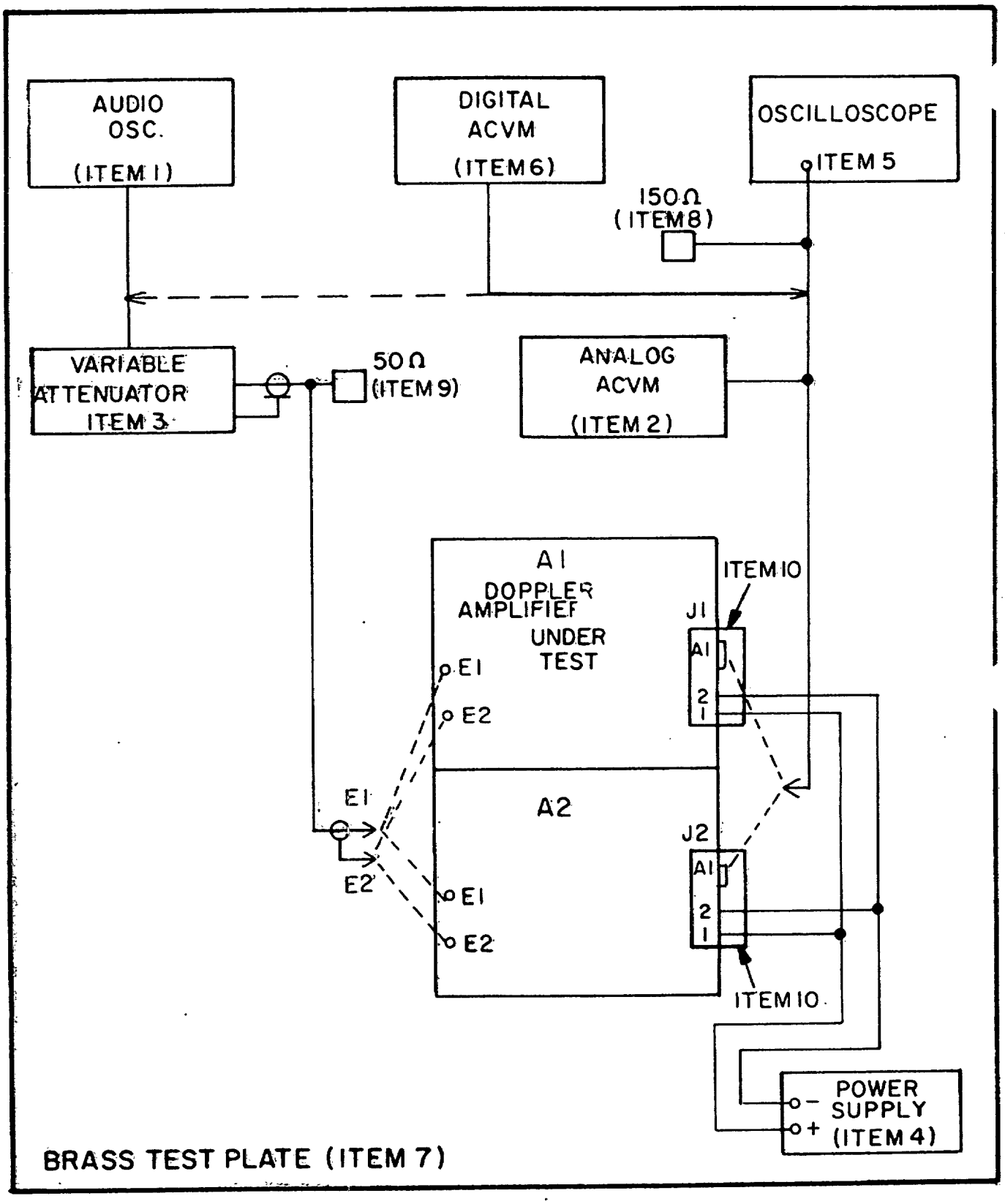


FIGURE 1. TEST EQUIPMENT SETUP

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4.6.3.3.2 Frequency Response. Monitor the output level at A1J1-A1 with the analog ac voltmeter, Item 2. While holding the output level at the audio oscillator, Item 1, constant at that voltage noted in 4.6.3.3.1, slowly vary its frequency between 900 Hz and 10 KHz while noting the highest and lowest output level of the UUT. The difference between the highest and lowest output level shall be 1.0 dB maximum. Increase the frequency to 20 KHz and note the output level at A1J1-A1. It shall be at least 2.0 dB below the highest level in the 900 Hz to 9 KHz range.

4.6.3.3.3 Limiting Action. With the input signal applied as in 4.6.3.3.1 use the oscilloscope, Item 5, to observe the top of one cycle of the output signal and decrease the variable attenuator setting, Item 3, until the top of the sine wave just starts to flatten out. The output signal at the point of limiting shall be 1.5 ±0.5 volts peak-to-peak (P-P). Remove 20 dB, then an additional 20 dB from the attenuator, Item 3, and observe that the output signal level does not increase above 2.0 V P-P in either case.

4.6.3.4 Amplifier A2. Repeat 4.6.3.3.1 thru 4.6.3.3.3. except use A2E1 and A2E2 (ret) for the input and A2J1-A1 for the output.

5. PACKAGING

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 11740261. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The doppler amplifier assembly is used in the Detector, Doppler 10548509 which is used in Radar Receiver R-1475/VPS-2, part of Radar Set AN/VPS-2. It is a broadband amplifier which provides ampification of low-level doppler signals and supplies them to a phase detector.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

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c. Selection of applicable level of preservation, packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

6.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring of the equipment. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAF-I, Dover, NJ 07801.

When the contractor submits inspection equipment designs to the Government for approval, he shall give the following information in his letter of transmittal:

a. The contract number.
b. The contract item (name, model number, etc.).
c. The designs remaining to be submitted and the expected date of transmittal.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.5 Distribution of original specification. MIL-A-48607(AR) was for internal US Army Armament Research and Development Command use only and was not forwarded to the Naval Publications and Forms Center for distribution.

6.6 Changes to previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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